WHICH TASKS MUST CHANGE TO WIN? EXACT SCORE-SPACE AUDITS OF MULTI-TASK AGGREGATION

Autonomous research report
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ABSTRACT

How much targeted improvement would make a model first on a multi-task leaderboard? We study this question as a conditional optimization problem on public score matrices, without estimating training responses. Investment is measured either by the number of improved tasks or by total score increase. We give exact algorithms for arithmetic mean, median, geometric mean, and minimum aggregation, including bounded and infeasible improvements. On 79 models across 57 HELM–MMLU subjects and 4496 Open LLM models across 6 evaluations, the apparent ordering of mean and median reverses with the improvement ceiling. With perfect scores allowed, median requires more changed tasks than mean for 83.1% and 47.4% of common challengers, respectively. With ceilings set to each task's best observed score, median requires fewer tasks for 97.4% and 62.5%. Yet median requires less total score increase for nearly every challenger. Task resampling and alternative score normalization further qualify these comparisons. Thus no aggregation rule has a universal investment advantage: the ceiling, cost definition, and task composition are part of the answer. These are reproducible score-space certificates, not estimates of data or compute needed to train a winning model.

1 INTRODUCTION

A multi-task leaderboard turns a vector of task scores into one rank. This creates two distinct questions: what abilities should count, and how easily could a developer change the winning rank by concentrating improvement on selected tasks? The latter matters when benchmark-aligned data curation or post-training improves measured performance without comparable improvements elsewhere. Experiments on benchmark leakage and rephrased test samples establish that this concern is substantive (Zhou et al., 2023; Yang et al., 2023). They do not identify the response of every model on every task, however. A score table alone cannot tell us how many training examples, tokens, or GPU hours would buy a given improvement.

We therefore ask a narrower, identifiable question. Conditional on a specified set of task-wise attainable scores, what is the least score-space intervention that makes a fixed model first while its rivals stay fixed? We measure intervention in two ways: the number of affected tasks and the average absolute increase across tasks. Both are useful audit quantities, but neither is a measured training cost. The first captures task-level setup or data-acquisition overhead; the second captures the amount of score movement. Their disagreement is itself a result.

One might expect a median to resist targeting because it ignores extreme scores. That intuition omits both the target's starting vector and the incumbent's threshold. An already strong model may need to move only one score across a median boundary. Conversely, a target with many scores just below that boundary may need numerous small changes, while a mean can be shifted through one large change. The familiar question of how many arbitrary observations destroy a location estimate is different from making one particular bounded vector beat the best competing vector.

Our main empirical finding is a reversal that holds on the same matrices and the same target cohorts. Allowing selected scores to reach one gives median a larger average task count than mean on both panels. Restricting ceilings to the best scores already observed in each task reverses that comparison for most targets. The second ceiling is a sensitivity scenario, not an assertion that borrowing another

model's best score is achievable. Both scenarios are deliberately explicit; treating either as ground truth about training would be unwarranted.

The contributions are threefold. First, we formulate minimum investment with fixed rivals, bounded improvements, ties, and unreachable winners. Second, we provide exact, transparent solvers and characterize when the same subset is optimal across symmetric aggregation rules. A constructive example shows how changing only the ceiling reverses the ordering. Third, we audit two public snapshots with shared challenger cohorts, alternative investment units, weaker responses, score-scale and row-selection controls, and task-composition perturbations. Our claims concern these audited settings and inspected literature; they do not assert exhaustive priority.

2 Related work

MMLU evaluates knowledge over many subjects (Hendrycks et al., 2021); HELM emphasizes standardized and transparent evaluation across scenarios and metrics (Liang et al., 2023). We use HELM's later MMLU score release and the public Open LLM Leaderboard table (Stanford CRFM, 2025; Open LLM Leaderboard, 2025). Our macro aggregation over HELM subjects is an explicit counterfactual scoring rule, not an attempt to reproduce every displayed HELM summary. The benchmark lottery shows that task choice changes relative algorithm performance (Dehghani et al., 2021); here task resampling tests whether an investment comparison survives a change of composition.

Lanctot et al. (2023) treat tasks as voters and study ordinal evaluation rules, including consistency properties and cycles. We examine four cardinal aggregators whose rivals' aggregate scores remain fixed when the target changes. Thus our threshold reduction does not directly cover Borda, pairwise tournaments, or other rules under which changing one model also changes competitors' scores. Election bribery asks whether bounded intervention can make a preferred candidate win (Faliszewski et al., 2006). Our intervention changes only the target's cardinal coordinates, with a box constraint rather than arbitrary ballot changes.

Chen et al. (2026) model a benchmark designer and multiple developers in a Stackelberg game and analyze equilibrium incentives under strategic post-training. We instead solve a unilateral, static, finite-table optimization problem; no equilibrium, latent ability, or post-training response is inferred. Contamination experiments (Zhou et al., 2023; Yang et al., 2023) motivate targeted improvement but also caution against treating independent, harmless score gains as an empirical law. These differences delimit our contribution rather than establish that the broader strategic phenomenon is new.

3 Problem and assumptions

Let $X \in [0,1]^{n\times d}$ contain complete, higher-is-better scores. For target i, write $x = X_i$ and let g be the aggregation rule. The competing threshold is

$$t_{i,g} = \max_{\ell \neq i} g(X_\ell) + \varepsilon. \tag{1}$$

Our primary convention is co-first place, $\varepsilon = 0$. A sensitivity analysis uses $\varepsilon = 10^{-6}$ on the unit score scale, a finite positive winning margin rather than an unattained strict-inequality infimum. Numerical comparisons use a 10^{-12} tolerance.

Feasible improvement. The target may choose any y with $x \le y \le u$. We parameterize

$$u_j = x_j + \rho(c_j - x_j), \qquad c_j = 1 \text{ (unit ceiling)}, \quad \text{or} \quad c_j = \max_\ell X_{\ell j} \text{ (observed ceiling)}, \tag{2}$$

where $\rho \in \{0.1, 0.25, 0.5, 0.75, 1\}$ specifies the fraction of available headroom. The unit ceiling supplies a deliberately optimistic limit. The observed ceiling keeps gains within the snapshot's empirical per-task envelope, but a vector combining different models' task maxima need not correspond to any trainable model. We do not fit ρ to data or interpret it as a learning rate.

Investment. We compute two separate optima:

$$K_g^*(x,u,t) = \min_{x \le y \le u} \{\|y-x\|_0 : g(y) \ge t\}, \tag{3}$$

$$C_g^*(x,u,t) = \min_{x \le y \le u} \{d^{-1}\|y-x\|_1 : g(y) \ge t\}. \tag{4}$$

An infeasible optimum is $+\infty$. K^* counts selected tasks, regardless of the size of their gains. C^* allows partial increases and is reported in average percentage points. The two optimizers may differ. Task costs are uniform: neither metric equates a 57-subject MMLU task with a whole Open LLM evaluation in training effort. Fractions K^*/d allow descriptive plots, not a conversion between those units.

Scope of the model. The developer knows all snapshot scores and the rule, rivals do not respond, gains on selected tasks can be controlled independently, and other scores do not change. These assumptions isolate the aggregation effect. They exclude transfer, forgetting, coupled learning curves, hidden tests, noisy score estimation, and multi-developer competition. Under these assumptions the unit-ceiling K^* is an optimistic lower bound for any smaller attainable box. It is not such a bound on the number of training domains if training on one domain also improves several other coordinates.

Aggregation rules. We study the arithmetic mean $d^{-1}\sum_j y_j$, the conventional median (averaging the middle pair when d is even), the geometric mean $(\prod_j y_j)^{1/d}$ with zero preserved, and the minimum $\min_j y_j$. The latter two emphasize weak tasks in different ways. All are symmetric and coordinatewise nondecreasing. We give no normative ranking of them: a minimum and a mean encode different notions of merit. Each rule uses its own best rival in Equation 1.

4 Exact investment certificates

The optimization is small enough to solve exactly for the two audited dimensions. The following results also explain why its answer depends on the intervention model. Proofs and continuous-cost algorithms are in Appendix A.

Proposition 1 (Endpoint reduction and a shared-ceiling optimum)**.** *For any coordinatewise nondecreasing g, a minimum-support feasible intervention can be taken to have $y_j \in \{x_j, u_j\}$. If $u_j = c \ge \max_j x_j$ for every j and g is also symmetric, raising the k smallest original scores to c maximizes $g(y)$ among all interventions on at most k coordinates.*

The second statement follows because replacing a larger original score instead of a smaller one leaves a smaller unchanged coordinate; exchanging the choice weakly increases every sorted coordinate. Consequently, at the full unit ceiling, "lowest score first" is an optimal strategy for all four rules. Their different budgets then reflect different winning boundaries, not different task-selection heuristics. With partial response or heterogeneous ceilings this conclusion need not hold.

Proposition 2 (Exact support costs)**.** *Let $\Delta = dt - \sum_j x_j$ and let $a_{(1)} \ge \cdots \ge a_{(d)}$ sort the gains $a_j = u_j - x_j$. For the mean,*

$$K^*_{\text{mean}} = \min\{k : \textstyle\sum_{r=1}^{k} a_{(r)} \ge \Delta\}. \tag{5}$$

The geometric mean has the same prefix form in log scores, after mandatory changes to every zero coordinate when $t > 0$. For an odd number $d = 2m-1$ of tasks, define $A = \#\{j : x_j \ge t\}$ and $B = \#\{j : u_j \ge t\}$. Then

$$K^*_{\text{med}} = \begin{cases} (m-A)_+, & B \ge m, \\ +\infty, & B < m. \end{cases} \tag{6}$$

For the minimum, every coordinate below t must be changed, and any $u_j < t$ makes winning infeasible. For any rule, feasibility with unrestricted support is equivalent to $g(u) \ge t$.

Mean and log-mean costs require sorting, $O(d \log d)$; odd median and minimum require counting, $O(d)$. On the six-task panel, all 2^6 endpoint subsets are enumerated for the even median. This avoids substituting a nonstandard lower or upper median for the official mathematical convention. The best feasible subset is retained as a witness.

Equation 6 also identifies a useful invariance: as long as at least m caps cross the median threshold, their exact magnitudes do not change K^*_{med}. The mean instead responds to the entire distribution of gains $u_j - x_j$. Reducing headroom can therefore change the mean's budget without changing the median's. In particular, at full response with co-first allowed, observed ceilings always permit the best rival's score vector. Thus odd-median and minimum support costs must equal their unit-ceiling costs; the even median has no such guarantee. No claim of a universal ordering is possible.

Proposition 3 (A ceiling-only reversal)**.** *There exists a single leaderboard, a fixed challenger, and fixed competing models such that $K^*_{\text{mean}} < K^*_{\text{med}}$ under the unit ceiling but $K^*_{\text{mean}} > K^*_{\text{med}}$ under the observed ceiling. This holds both for co-first place and for a sufficiently small positive winning margin.*

For a concrete proof construction, take target $(.59, .59, .59)$, incumbent $(.60, .60, .60)$, and three specialist rivals whose only nonzero coordinate is $.604$, one on each task. Both thresholds are $.60$. Unit ceilings give budgets $(K^*_{\text{mean}}, K^*_{\text{med}}) = (1, 2)$; observed ceilings give $(3, 2)$. These are algebraic counterexamples, not empirical benchmark observations. Appendix A also gives an example with median cheaper than mean at the unit ceiling.

For continuous investment, the mean has $C^*_{\text{mean}} = (t - g_{\text{mean}}(x))_+$ whenever feasible. The odd median sums the m smallest feasible deficits $(t - x_j)_+$, including zero deficits, then divides by d. The minimum sums all deficits. The geometric solution raises low coordinates to a common level, subject to their caps, as derived from the log constraint. An exact small-dimensional algorithm for the even median is given in Appendix A.4. There is no reason for the ordering in task count to match that in score mass.

5 Empirical audit

5.1 Data, targets, and validation

HELM–MMLU. We use release v1.13.0, dated January 10, 2025, downloaded November 2, 2026. Its subject JSON summaries yield 79 evaluated model entries with all 57 subject exact-match scores. Model/protocol identifiers are taken from run-spec metadata, not display labels. No entry is dropped for missing scores. Subjects receive equal weight in the primary audit; evaluation-instance weighting is a separate control (Appendix B).

Open LLM. We use the March 20, 2025 contents snapshot pinned to commit `9c09a7c...afa2`. The downloaded table has 4576 rows over IFEval, BBH, MATH level 5, GPQA, MuSR, and MMLU-Pro. We remove 1 flagged entry, then choose one record per model name by a score-independent precision priority and lexicographic evaluation identifier, removing 79 extra records. The resulting 4496 models are complete on all six tasks. We divide displayed normalized scores by 100; their equal-weight mean exactly reproduces the provided average. Raw metric columns and all unflagged evaluation entries are separate sensitivity panels. We do not assume that heterogeneous metrics have a shared scientific utility merely because they share a numeric range.

For paired rule comparisons we fix a *common challenger cohort*: entries that are not initially co-first under any of the four rules. This leaves 77 HELM and 4493 Open LLM targets. Rival sets still contain all retained entries, including initial winners. Thus no rule gains an easier target sample by excluding its own losers. The same cohort is used across ceilings, response strengths, and winning margins. A top-20% cohort uses original mean rank, with the same cross-rule exclusion. All snapshots, checksums, row-selection rules, and model-level results accompany the artifact.

All calculations are deterministic except explicitly seeded perturbations and random-order baselines. Small matrices are checked against exhaustive subset search; median mass is checked against independently formulated linear programs over score orderings, and geometric mass against constrained numerical optimization. Across 109800 primary-panel cases we audit endpoint witnesses when feasible. Optimality on the full panels follows from the formulas or full enumeration, rather than a heuristic solver. The main run used 111.9 CPU seconds and 378.4 MiB peak resident memory.

Panel	Ceiling	Mean	Median	Geom.	Min.	$K_{\text{med}} < K_{\text{mean}}$ (%)	$>$ (%)
HELM–MMLU	Unit	17.12	19.82	17.35	13.52	14.3	83.1
HELM–MMLU	Observed	34.96	19.82	34.13	13.52	97.4	0.0
Open LLM	Unit	2.36	2.85	2.98	3.66	0.0	47.4
Open LLM	Observed	3.88	3.24	4.65	3.66	62.5	0.2

Table 1: Mean minimum task count K^* at $\rho = 1$, co-first convention. The final columns report the fraction of common challengers for which median costs fewer or more tasks than mean. Equal cases make up the remainder. Each panel uses its own task unit; counts should not be compared as training costs.

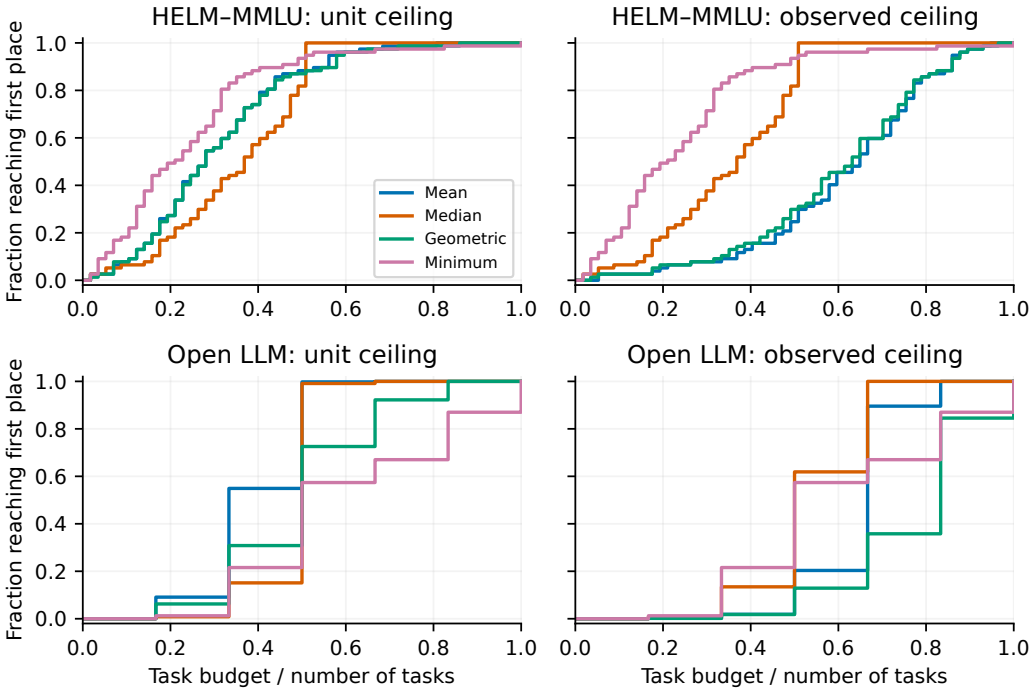


Figure 1: Exact fraction of common challengers that can reach first place within each task budget at $\rho = 1$. Crossings and ceiling dependence prevent a uniform rule ranking. The steps reflect discrete task counts, especially on six-task Open LLM. All targets reach co-first by the full budget in these scenarios.

5.2 The ceiling reverses the task-count comparison

Table 1 reports finite-census averages of exact minimum task counts; all targets are feasible at $\rho = 1$. Under unit ceilings, mean needs fewer tasks than median for 83.1% of HELM challengers and 47.4% of Open LLM challengers. Median is cheaper for 14.3% and 0.0%, respectively. Under observed ceilings, median becomes cheaper for 97.4% and 62.5%. These are paired comparisons for the same models, not comparisons of independently averaged rank groups.

HELM makes the mechanism particularly clear. Changing from unit to observed ceilings increases average mean cost from 17.12 to 34.96 tasks, while median remains at 19.82. Enough coordinates can still cross its threshold, while the caps reduce the aggregate mass obtainable per changed coordinate. Open LLM's even median also becomes more expensive, from 2.85 to 3.24, but mean increases more, from 2.36 to 3.88.

Figure 1 shows why average budgets do not establish a full dominance claim. On HELM with unit ceilings, the median can be easier for some large-budget targets even though it is harder on average. The minimum has the lowest average task count on HELM, but a larger average than mean on Open

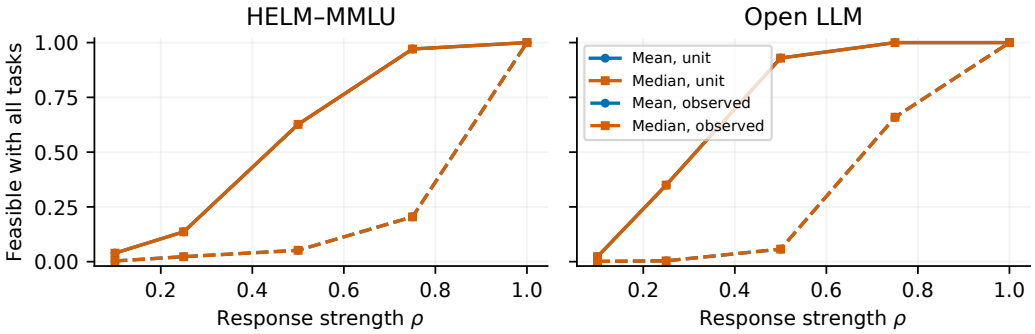


Figure 2: Fraction of the fixed common challenger cohort for which winning is feasible with all tasks available. Solid lines use unit ceilings; dashed lines use observed ceilings. A larger task budget cannot rescue a target whose full attainable vector remains below the incumbent.

Panel	Ceiling	Mean	Median	Geom.	Min.	$C_{\text{med}} < C_{\text{mean}}$ (%)
HELM–MMLU	Unit	14.56	3.44	14.10	3.09	100.00
HELM–MMLU	Observed	14.56	3.72	14.29	3.09	100.00
Open LLM	Unit	30.25	18.44	25.79	8.52	99.93
Open LLM	Observed	30.25	18.81	26.99	8.52	99.93

Table 2: Mean minimum *average score increase* $100C^*$, in percentage points, at $\rho = 1$. The final column is a paired comparison of score mass. This is a different optimization from saturating a minimum number of tasks.

LLM (Table 1). Its incumbent is defined by the worst task, so enforcing that lower winning threshold can be cheap. Penalizing weak performance does not by itself establish resistance to a change in the winning rank.

Targeted task selection is a consequential baseline control under heterogeneous caps. Under unit ceilings, lowest-first and optimal costs coincide, as predicted by Proposition 1. Under observed ceilings on HELM, lowest-first requires 43.31 tasks on average for median, versus the optimum 19.82. The gap arises because spending on the weakest scores can fail to cross the incumbent's high median boundary. Random ordering is also substantially more costly (Appendix C); it cannot establish a lower bound on an informed developer's investment.

5.3 Weaker improvements can make winning impossible

A small task count under an optimistic ceiling does not imply a reachable training outcome. Figure 2 gives feasibility even when every task may be changed. With unit ceilings at $\rho = .5$, the feasible fractions on Open LLM are 96.8% for mean and 75.0% for median. Under observed ceilings they fall to 3.5% and 10.1%. On HELM, observed-ceiling feasibility at the same response is 2.6% and 6.5%. These failures are exact consequences of $g(u) < t$, not optimization timeouts. We report feasibility rather than assign finite effort to such cases. Because neither ceiling nor ρ is learned from training interventions, the audit cannot select which of these regimes describes an actual developer.

5.4 Counting tasks and counting score increases give different answers

Table 2 changes the investment unit while keeping scores, rivals, and thresholds fixed. On unit-ceiling HELM, median needs less score mass than mean for 100.00% of targets, even though it needs more changed coordinates for 83.1%. Its average minimum increase is 3.44 percentage points per task versus 14.56 for mean. On Open LLM the corresponding values are 18.44 and 30.25, with median cheaper for 99.93% of targets. Thus interpreting task count as total training effort could reverse an audit's practical message.

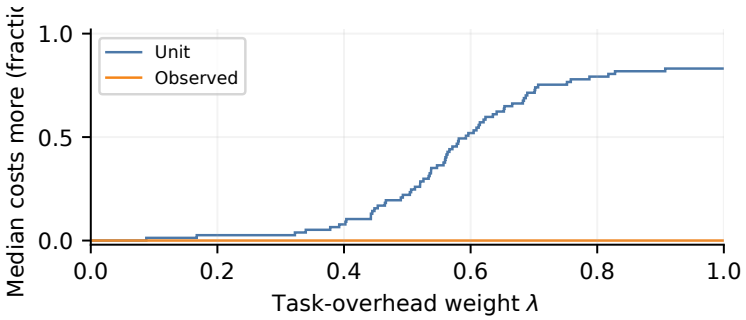


Figure 3: Exact mixed-cost comparison on HELM at $\rho = 1$. The vertical axis is the fraction of common challengers with $J^*_{\lambda,\mathrm{med}} > J^*_{\lambda,\mathrm{mean}}$. A larger overhead weight can reverse the unit-ceiling comparison, while observed-ceiling median costs never exceed mean costs here.

The mean's mass optimum is exactly its gap to the best rival and is unchanged by the ceiling whenever feasible. The odd median instead pays just the cheapest deficits needed to populate its upper half. It can therefore be insensitive to many very low scores. The minimum ignores no deficit below its own winning threshold, yet that threshold can be much lower than a mean or median threshold. The empirical mass distributions are shown in Appendix D. Neither investment unit includes task-specific acquisition difficulty or a cost for changing a model's representation.

An exact overhead–gain tradeoff. For the mean and the odd-dimensional median, there is an intervention that simultaneously attains minimum support and minimum score mass (Appendix A.5). Thus a cost with normalized task overhead

$$\begin{aligned} J^*_{\lambda,g} &= \min_{x \le y \le u,\ g(y) \ge t} \left\{ \lambda \frac{\|y - x\|_0}{d} + (1-\lambda) \frac{\|y - x\|_1}{d} \right\} \\ &= \lambda \frac{K^*_g}{d} + (1-\lambda) C^*_g \end{aligned} \tag{7}$$

can be computed exactly from our certificates. This identity is not asserted for an even median or the geometric mean. On unit-ceiling HELM, 64 challengers switch from median cheaper at $\lambda = 0$ to median costlier at $\lambda = 1$. Their crossover weights have median 0.563 and interquartile range $[0.492, 0.644]$. Under observed ceilings median is never more expensive for any weight in this cohort (Figure 3). The weights are hypothetical cost ratios, not fitted training parameters. They expose which notion of investment is responsible for a robustness claim.

5.5 Sensitivity and what uncertainty means here

These tables are censuses of two retained snapshots, so their optimization results have no Monte Carlo estimation error. The central uncertainties concern what counts as a task, which models are represented, and what gains training can attain. We expose the first two through two reproducible perturbations. We resample owner groups of targets, using the model namespace as a coarse cluster, and compute 2,000 paired reweightings. Separately, we draw 300 task multisets with replacement, preserve their original dimension, recompute all competing thresholds and caps, and retain the original target cohort. Table 3 reports their central 95% ranges for the mean paired difference $\Delta K = K^*_{\mathrm{med}} - K^*_{\mathrm{mean}}$. These are sensitivity ranges, not confidence intervals for independent models, unseen tasks, or true test accuracy.

Owner reweighting preserves the signs of all four average differences in these ranges. Task perturbation is less stable: both unit-ceiling intervals cross zero, as does the observed-ceiling Open LLM interval. The observed-ceiling HELM contrast is more stable within this particular perturbation. Repeated heterogeneous tasks should be interpreted as a change of benchmark emphasis, not a plausible sample of newly collected evaluation questions.

The main direction of the ceiling reversal persists among the top 20% by original mean rank and when retaining all unflagged evaluation entries. Moving Open LLM from displayed normalized columns to

Panel	Ceiling	$100\overline{\Delta K/d}$	Owner reweighting	Task resampling
HELM–MMLU	Unit	4.74	[2.19, 6.96]	[-0.07, 11.14]
HELM–MMLU	Observed	-26.57	[-28.61, -24.71]	[-34.03, -21.17]
Open LLM	Unit	8.12	[6.93, 9.26]	[-2.30, 15.72]
Open LLM	Observed	-10.70	[-11.42, -9.99]	[-32.46, 2.71]

Table 3: Paired task-count difference as a percentage of panel dimension. Positive means median costs more. Owner reweighting and task resampling are different perturbations of a finite table, not statistical evidence about a random population.

raw metric columns reduces the observed-ceiling fraction with median cheaper from 62.5% to 11.0%: most cases then tie in task count. This is a substantive scale limitation, not evidence for normalization invariance. A positive winning margin changes some HELM budgets but preserves the reversal; no Open LLM budget changes in this check. Evaluation-instance weighting also changes HELM mean budgets. Full controls and per-rule incumbents appear in Appendix B.

6 Discussion and limitations

The answer to "how much investment is needed?" is a conditional frontier, not one number. A leaderboard designer can publish (K^*, C^*) against explicit attainability scenarios, show where winning is impossible, and inspect witnesses near incumbent boundaries. The designer must still choose a scientific objective. A rule that makes rank change difficult may merely encode a different notion of quality or a low bottleneck threshold. Our results do not recommend minimum, mean, or median as an overall evaluation policy.

The principal limitation is identification. Public scores contain no intervention data mapping domain-specific training examples to gains, spillovers, or forgetting. Observed task maxima are attainable by some entry individually, not necessarily jointly or by the target. Saturation can overstate what post-training achieves; no negative transfer can understate its cost. Unequal task sizes and metrics make uniform task count and uniform score mass incomplete proxies. Even continuous score changes idealize the discrete increments of finite accuracy tests. Our formulas concern reported scores, and we cannot attach test-item confidence intervals without corresponding example-level outputs and a defensible sampling model.

The data are historical, selected, and contain correlated model families. Owner clusters imperfectly capture shared lineage; the small number of heterogeneous Open LLM task groups limits any task-general claim. A task-group attack may mean improving many subtasks, whereas a HELM coordinate is one subject. Ordinal aggregators, hidden evaluations, and responding rivals require other optimization models. The literature protocol was intentionally restricted; no exhaustive related-work or novelty certification is claimed. None of the computed witnesses establishes contamination by any named model, and no model was trained here.

7 Conclusion

Exact score-space certificates make the required assumptions visible. On two public leaderboards, changing only the improvement ceiling reverses the average task-count comparison of mean and median; changing the investment unit changes it again. The same analysis also identifies unreachable winners and sensitivity to benchmark composition. Claims about aggregation and strategic investment should therefore specify the attainable gains, winning convention, task unit, and cost measure.

AI USE STATEMENT

This report was produced by an autonomous Codex research agent in response to a user-specified task. AI was used for conceptual modeling, mathematical statements and proofs, hypothesis formulation, experimental design, implementation, data cleaning, quantitative analysis, interpretation, literature selection and reading, figures, and all manuscript drafting and editing. No benchmark examples or empirical scores were generated by an AI model. The numerical results come from executed code on downloaded public score tables. Automated validation includes exhaustive searches, independently formulated optimization checks, and endpoint witness audits. This is not independent human review, and no human verification of the paper is claimed. No models were trained, and no paid inference API was used for experiments. The report is a research deliverable, not a submission to or publication at ICLR.

ETHICS STATEMENT

The study uses public aggregate evaluation scores without private user data or human-subject experiments. No benchmark was contaminated and no evaluated model was modified. Minimum-investment witnesses describe potential score sensitivity; they are not allegations against developers. Such audits can help benchmark maintainers identify fragile incentive structures, while also revealing how a known aggregation rule could be targeted. We release score-space analysis rather than training or data-leakage pipelines. Data-source cards do not declare a separate score-data license; the artifact records this uncertainty and makes no claim that the underlying models' licenses license leaderboard data.

REPRODUCIBILITY STATEMENT

The artifact contains the LaTeX source, the official unmodified ICLR 2027 style, pinned dependencies, frozen score summaries, SHA-256 checksums and source URLs, complete processing and experiment scripts, model-level results, and generated tables and plots. Equations 3–6 and Appendix A specify the algorithms. Appendix E and `README.md` give exact commands, seeds, runtime, and the table/figure map. The access protocol in `RESEARCH_LOG.md` records the restricted literature review, failed URL probes, and development runs. A completed artifact checklist appears in Appendix F; the inspected ICLR instructions require the AI statement above, rather than a NeurIPS-style paper checklist.

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A PROOFS AND EXACT ALGORITHMS

A.1 ENDPOINT REDUCTION AND SHARED CEILINGS

Proof of Proposition 1. For a feasible y, let $S = \{j : y_j > x_j\}$. Set $z_j = u_j$ for $j \in S$ and $z_j = x_j$ otherwise. Then $z \geq y$ coordinatewise, hence $g(z) \geq g(y)$, and its support has size at most $|S|$. This proves endpoint sufficiency without convexity or symmetry. If all endpoints equal c, consider a selected coordinate with original value b and an unselected coordinate with original value $a \leq b$. The resulting pair is (a, c). Exchanging selection gives (b, c), whose sorted coordinates are no smaller. After inserting the other unchanged coordinates, the entire sorted vector is also coordinatewise no smaller. Symmetry allows us to compare these sorted vectors, and monotonicity prevents a decrease in g. Repeating exchanges selects the k smallest original values. An intervention on fewer than k coordinates can be extended without decreasing its value. Thus the result holds for "at most k" as well. □

This result does not cover unequal task weights, heterogeneous caps, or a common fractional response $\rho < 1$. In the last case the endpoints increase with the original scores, and an exchange changes both sorted endpoints. For the mean and geometric mean under a unit ceiling, sorting gains nevertheless remains exact at every response strength.

A.2 SUPPORT FORMULAS, ZEROS, AND INFEASIBILITY

Proof of Proposition 2. For mean aggregation, saturation on subset S increases the unnormalized sum by $\sum_{j \in S}(u_j - x_j)$. Among size-$k$ subsets this is maximized by the largest k gains, proving Equation 5. The empty prefix is allowed, and no prefix large enough means infeasibility. For the geometric mean with positive coordinates, use $\sum_j \log y_j \geq d \log t$ and gains $\log u_j - \log x_j$. If $t > 0$, every zero original coordinate must be selected and must have $u_j > 0$. Replace these zeros by their caps to form a base log sum, then sort the log gains of the remaining positive coordinates. Add the mandatory zero count to the required prefix length. If any zero has zero cap, winning is impossible. For $t \leq 0$, a nonnegative score vector is already feasible.

For $d = 2m - 1$, its median is at least t precisely when at least m coordinates are at least t. Exactly A already satisfy this, and at most B can ever satisfy it. Any $m - A$ eligible coordinates below t suffice when $B \geq m$. The minimum is at least t if and only if every coordinate is at least t, which gives its formula. Finally, monotonicity implies that $g(u)$ is the maximum attainable score; hence $g(u) \geq t$ is necessary and sufficient for feasibility. □

At $\rho = 1$, $\varepsilon = 0$, and $u_j = \max_\ell X_{\ell j}$, the best rival's vector is coordinatewise bounded by u. An odd-median rival at score t has at least m coordinates at or above t, ensuring $B \geq m$; its support cost is then $(m - A)_+$ under both observed and unit ceilings. For the minimum, all observed caps are at least the best rival's minimum, so the same original deficits must be changed under either ceiling. This establishes the co-first invariances stated in Section 4. It does not assert equality of continuous median costs, nor feasibility with a positive margin above that rival.

The even median is not replaced by either central order statistic. The code enumerates all 64 subsets when $d = 6$, sorts their improved score vectors, averages the central pair, and returns the minimum feasible subset cardinality. This is an exact exponential algorithm restricted to a small dimension, not a claimed polynomial algorithm for arbitrary even d. Weighted means are also exact: sort gains $w_j(u_j - x_j)$ and cover $t - \sum_j w_j x_j$, where $\sum_j w_j = 1$.

A.3 NO UNIFORM ORDERING AND A CEILING REVERSAL

Proof of Proposition 3. Use the construction in the main text. Each specialist has mean $.604/3 < .60$ and median zero, so the uniform incumbent sets both thresholds. With unit caps, one changed target coordinate raises its mean to $(1 + .59 + .59)/3 > .60$, while its median remains .59. Two changes suffice for the median. With observed caps $.604$ on each coordinate, two changes give a mean $(.604 + .604 + .59)/3 < .60$; three give .604. Two changes give median .604. These strict comparisons persist at a 10^{-6} margin, as also checked in code. Only the cap model has changed. The target, rivals, rules and task units are fixed. □

The reverse task-count ordering can already occur at a unit ceiling: against incumbent $(.6, .6, .6)$, a target $(0, .1, .61)$ needs two changed tasks for mean but only one for median. A single mean improvement yields at most $(1 + .1 + .61)/3 = .57$, whereas changing the $.1$ coordinate makes the median $.61$. These examples rule out deterministic dominance over all score tables. They are mathematical proof constructions, explicitly excluded from every empirical table and figure.

A.4 Continuous score-mass investment

Assume the target is below threshold and that the endpoint vector is feasible. For the mean, the required total gain is $dt - \sum_j x_j$. A box allows every total gain between zero and $\sum_j (u_j - x_j)$, so this gain is both necessary and attainable. Dividing by d gives the mean formula. For the minimum, every deficit $(t - x_j)_+$ must be paid; summing these independent necessities is attainable whenever all caps cross the threshold.

For the odd median, assign cost $b_j = (t - x_j)_+$ to any coordinate with $u_j \geq t$ and $b_j = +\infty$ otherwise. At least m coordinates must reach t, and each has exactly that minimum cost. Therefore

$$C^*_{\mathrm{med}} = d^{-1} \sum_{r=1}^{m} b_{[r]}, \tag{8}$$

where $b_{[r]}$ are sorted increasingly. Zero costs account for coordinates already above threshold. Infeasibility is retained if fewer than m finite costs exist.

For the geometric mean at $t > 0$, minimize $\sum_j (y_j - x_j)$ subject to the box and $\sum_j \log y_j \geq d \log t$. This is a convex feasible region. The Lagrangian stationarity condition on an interior coordinate is $1 - \lambda / y_j = 0$. Hence

$$y_j = \min\{u_j, \max\{x_j, \lambda\}\}, \qquad \sum_j \log y_j = d \log t. \tag{9}$$

Monotonicity gives a one-dimensional root; the code uses Brent's method with absolute root tolerance 10^{-14}. Clamped flat segments do not change the minimum attained mass. An infeasible endpoint or a zero cap at a positive threshold is handled before root finding.

For an even dimension $d = 2m$ (six in our data), median feasibility means that $y_{(m)} \geq a$ and $y_{(m+1)} \geq b$ for some $a \leq b$ with $a + b = 2t$. Choose a set H of m coordinates to reach b and one additional coordinate $\ell \notin H$ to reach a. The others stay unchanged. For this assignment, the total cost as a function of a is

$$f_{H,\ell}(a) = \sum_{j \in H} (2t - a - x_j)_+ + (a - x_\ell)_+, \tag{10}$$

on the interval

$$\max\{0, 2t - \min_{j \in H} u_j\} \leq a \leq \min\{t, u_\ell\}. \tag{11}$$

Empty intervals are infeasible. Otherwise this piecewise-linear convex function attains a minimum at an endpoint or a breakpoint $a = x_\ell$ or $a = 2t - x_j$. We evaluate those clipped candidates and minimize over all (H, ℓ). For necessity, take the top m coordinates and the next coordinate of any feasible y, set $a = y_{(m)}$ and $b = 2t - a \leq y_{(m+1)}$, and reduce unnecessary increases. If $y_{(m)} > t$, one may instead set $a = b = t$. Thus the enumeration covers an optimum. No relaxation is used.

A.5 Joint attainment for mixed overhead and score cost

For a feasible mean threshold above the original score, choose a minimum-support set of largest headrooms. Its total headroom covers $dt - \sum_j x_j$, whereas every smaller set fails to cover that gap. Allocate exactly that total gain over the chosen coordinates. Such an allocation exists in the box; if any chosen coordinate had zero gain, a smaller support would cover the gap, a contradiction. This intervention attains both K^*_{mean} and C^*_{mean}. For an odd median, write $K = m - A > 0$. Choose the K smallest positive deficits among coordinates whose caps reach t, and raise them exactly to t. Together with the A already qualifying coordinates, this attains both the minimum support and the minimum mass. If the target already wins, the zero intervention attains both. Any nonnegative

weighted sum of two separately minimized costs is at least the corresponding weighted sum of their minima; these common optimizers attain equality, proving Equation 7. This proof requires neither shared caps nor full saturation, but does require feasibility. The figure uses the feasible $\rho = 1$ HELM cohort. When $\Delta s = (K^*_{\text{med}} - K^*_{\text{mean}})/d > 0$ and $\Delta c = C^*_{\text{med}} - C^*_{\text{mean}} < 0$, the crossover is exactly $\lambda_* = -\Delta c/(\Delta s - \Delta c)$.

B Data and experimental controls

B.1 Acquisition, completeness, and score units

The download script requests the frozen HELM release listing, its release summary, and exactly the 57 files matching `groups/json/mmlu_mmlu_subject:*.json`; similarly named alternative group tables are not duplicated. Each subject table has one run specification per row, from which the model identifier is parsed. We extract its `EM` column and retain evaluation counts for the micro-mean control. All 79 identifiers occur in every subject. The Open LLM parquet and dataset card are downloaded at the full commit in `data/manifest.json`. In total the manifest covers 61 raw files, 8.97 MB. No full run-output collection or evaluation-set examples are needed.

The Open LLM precision priority is bfloat16, float16, float32, 8bit, 4bit, then other labels. Ties use lexicographic `eval_name`; the first row per `fullname` is kept. This rule consults no scores. The full-entry sensitivity keeps all complete, unflagged `eval_name` records. The raw-score sensitivity uses the same selected rows and the six `Raw` columns without dividing them by 100, since those columns are already on the unit scale. These two scales have different task-specific baselines and are not related by a common monotone transformation of the aggregate. Macro scores are carried to CSV with 17 significant digits.

Panel	Rule	Maximum	Entry / model
HELM–MMLU	Mean	0.872730	anthropic_claude-3-5-sonnet-20241022
HELM–MMLU	Median	0.898148	anthropic_claude-3-5-sonnet-20241022, deepseek-ai_deepseek-v3
HELM–MMLU	Geometric	0.867504	deepseek-ai_deepseek-v3
HELM–MMLU	Minimum	0.596386	deepseek-ai_deepseek-v3
Open LLM	Mean	0.520814	MaziyarPanahi/calme-3.2-instruct-78b
Open LLM	Median	0.556164	ehristoforu/qwen2.5-test-32b-it
Open LLM	Geometric	0.472896	MaziyarPanahi/calme-3.2-instruct-78b
Open LLM	Minimum	0.231464	rootxhacker/Apollo-70B

Table 4: Initial competing thresholds and winners under each audited rule. HELM median has a tie. These entries remain in every target's rival set.

The original Open LLM score table contains model hub-license metadata but no dataset-level license declaration. Neither the inspected HELM summaries nor the public release listing states a separate score-data license. Access without an account does not establish an open-data license, and HELM code's Apache license is not assumed to apply to these scores. This artifact uses aggregate scores for reproducibility and records sources and checksums, without asserting third-party redistribution rights. A fresh-download path is provided.

B.2 Alternative target cohorts and normalization

The top-cohort control preserves the direction of the primary mean–median comparison for both ceilings. The raw-column observed-ceiling comparison is much closer to a tie; results on one normalization should not be generalized to another. Full-entry retention changes the fractions only slightly in this snapshot but does not eliminate the more general problem of dependent model submissions.

Panel / cohort	Ceiling	Targets	Mean K	Median K	Median cheaper (%)
HELM–MMLU top 20%	Unit	14	5.71	7.71	14.3
HELM–MMLU top 20%	Observed	14	17.43	7.71	85.7
Open LLM top 20%	Unit	897	1.54	2.26	0.0
Open LLM top 20%	Observed	897	3.03	2.29	71.3
Open LLM (raw)	Unit	4494	2.35	2.93	0.0
Open LLM (raw)	Observed	4494	3.66	3.61	11.0
Open LLM (all entries)	Unit	4572	2.36	2.85	0.0
Open LLM (all entries)	Observed	4572	3.88	3.24	62.4

Table 5: Exact $\rho = 1$ task-count controls. "Top 20%" means mean rank at most $\lceil .2n \rceil$, then common nonleader exclusion. Raw and full-entry panels each recompute incumbents and their own common cohort.

Panel	Ceiling	Any K changed	Mean K	Median K	Median cheaper (%)
HELM–MMLU	Unit	7	17.12	19.90	14.3
HELM–MMLU	Observed	7	34.96	19.90	97.4
Open LLM	Unit	0	2.36	2.85	0.0
Open LLM	Observed	0	3.88	3.24	62.5

Table 6: Positive-margin control ($\varepsilon = 10^{-6}$), $\rho = 1$. "Any K changed" counts targets whose support budget changes for at least one of the four rules relative to co-first. All remain feasible.

B.3 Winning margins and subject weighting

The denominator for this weighting control differs from the main cohort because only the two mean variants are compared. Task counts still assign unit overhead to each subject, even when score contributions are weighted by subject size.

C Strategy baselines

For a random permutation and budget k, its first k tasks are uniformly distributed over size-k subsets. If p_k is the fraction of such subsets that win, monotonicity yields $\mathbb{E}K = \sum_{k=0}^{d-1}(1 - p_k)$ whenever the full budget is feasible. We enumerate those fractions on the six-task panel. For HELM, permutation draws estimate this expectation; they are a randomized baseline, not part of the exact optimum. The artifact also records $\rho = .5$ baseline outcomes, but does not average their infeasible codes as finite effort. The full model-level outputs allow different budget thresholds to be inspected.

D Additional distributions

The bootstrap is used only as a controlled perturbation of benchmark composition. It does not repair missing training-response information. In particular, replicating one of six heterogeneous Open LLM groups can substantially change the scientific meaning of the benchmark. This is precisely why the broad observed-ceiling comparison on that panel is weaker than a universal robustness claim.

E Reproduction details

The environment is Python 3.11.2, NumPy 2.2.6, pandas 2.3.3, SciPy 1.15.3, Matplotlib 3.10.7, PyArrow 22.0.0, and requests 2.32.5. Full transitive versions are in `requirements-lock.txt`; TeX uses the official ICLR 2027 style downloaded from the URL in its author guidelines. The style files are unchanged. Only the running header is set in the manuscript to avoid suggesting publication or an actual conference submission. We enforce the initial-submission nine-page main-text ceiling. AI, ethics, and reproducibility statements, references, and appendices are outside that limit according to the inspected instructions.

Ceiling	Targets	Macro mean K	Micro mean K	Different K (%)
Unit	78	16.91	11.27	96.2
Observed	78	34.53	33.74	75.6

Table 7: HELM arithmetic mean with equal subject weights (macro) versus weights proportional to evaluated question counts (micro). Cohort: entries not initially co-first under either mean, rather than the four-rule cohort of the main audit.

Panel	Ceiling	Rule	Optimal	Lowest first	Random order
HELM–MMLU	Unit	Mean	17.12	17.12	27.94 ± 0.01
HELM–MMLU	Unit	Median	19.82	19.82	22.48 ± 0.01
HELM–MMLU	Unit	Geometric	17.35	17.35	30.71 ± 0.02
HELM–MMLU	Unit	Minimum	13.52	13.52	51.43 ± 0.04
HELM–MMLU	Observed	Mean	34.96	36.53	46.69 ± 0.02
HELM–MMLU	Observed	Median	19.82	43.31	44.15 ± 0.03
HELM–MMLU	Observed	Geometric	34.13	35.26	47.41 ± 0.02
HELM–MMLU	Observed	Minimum	13.52	13.52	51.35 ± 0.04
Open LLM	Unit	Mean	2.36	2.36	2.74
Open LLM	Unit	Median	2.85	2.85	3.02
Open LLM	Unit	Geometric	2.98	2.98	3.95
Open LLM	Unit	Minimum	3.66	3.66	5.36
Open LLM	Observed	Mean	3.88	4.84	4.82
Open LLM	Observed	Median	3.24	5.23	5.18
Open LLM	Observed	Geometric	4.65	4.70	5.31
Open LLM	Observed	Minimum	3.66	3.66	5.36

Table 8: Mean number of tasks before reaching co-first, $\rho = 1$. Lowest-first orders by original task scores. Random-order expectation is exact over subsets on Open LLM and estimated using 256 independent random permutations per target on HELM (seed 20261102); $\pm$ gives one Monte Carlo standard error of the finite-cohort mean, propagating independent target-wise draw variances. The optimal and lowest-first entries coincide at unit ceilings as a theorem, not a numerical coincidence.

From the artifact root, create a virtual environment, install the pinned requirements, and run the following commands:

```
python -m src.fetch_data
python -m src.prepare_data
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
  python -m src.run_experiments --task-bootstrap 300
OPENBLAS_NUM_THREADS=1 python -m src.report
python -m pytest -q
cd paper && latexmk -pdf -interaction=nonstopmode main.tex
```

The download step verifies existing frozen files against the manifest instead of silently replacing them. Internet access is only needed for a fresh download. Processing, experiments, plots and PDF compilation can be rerun offline with the included raw scores. The recorded main run takes 112.7 seconds wall time and 111.9 process CPU seconds with BLAS/OpenMP limited to one thread; actual speed varies by host. It uses 378.4 MiB peak resident memory. No GPU, model execution, or paid API is needed.

The master seed is 20261102 for random-order baselines, 20261103 for task multisets, and 20261104 for owner-group reweighting. Exact optimization itself has no random seed. Coarse HELM owners use the identifier prefix before the first underscore; Open LLM owners use the namespace before the slash. In owner reweighting, entire groups are sampled with replacement, and both numerator and denominator of the paired target mean are reweighted together. This retains within-group dependence as represented by that imperfect proxy.

Every paper table and figure is generated by `src/report.py` from `artifacts/*.csv`. Source scores and optimization results come from `prepare_data.py` and `run_experiments.py`. Specifically, Table 1 uses `support.csv`; Table 2 and Figure 4 use `mass.csv`; Table 3 and

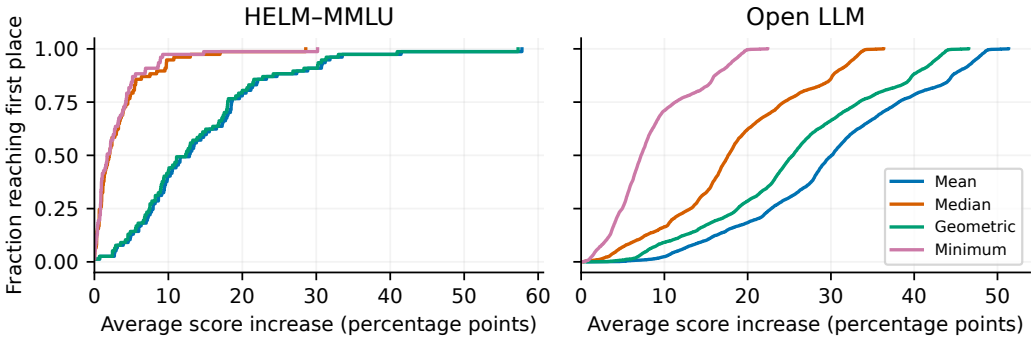


Figure 4: Exact continuous score-mass frontiers under unit ceilings at $\rho = 1$. The horizontal unit is average percentage-point increase across a panel's tasks, not training examples or compute. These curves solve Equation 4.

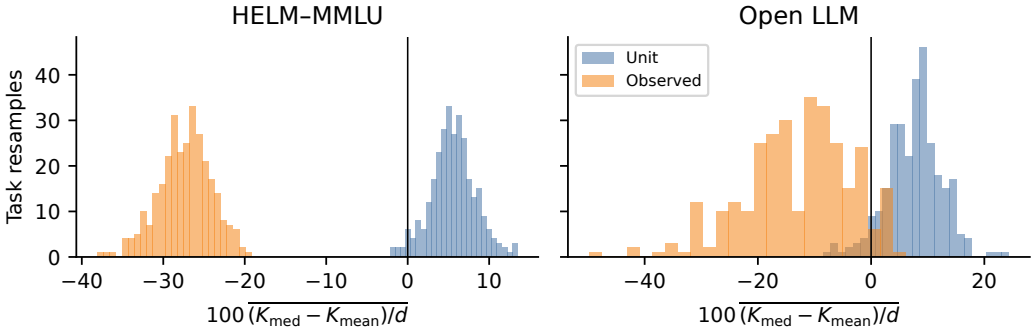


Figure 5: Mean paired support difference over 300 task multisets per panel. Within a draw, all models share the same sampled tasks and thresholds and caps are recomputed. The original common target cohort is held fixed. Crossing zero indicates that changing task emphasis can change the comparison.

Figure 5 additionally use `task_bootstrap.csv` and namespace metadata. Figures 1 and 2 and Tables 5 and 6 use `support.csv`. Table 4 aggregates processed matrices directly; Table 7 uses `weighting.csv`; Table 8 uses `baselines.csv` and `support.csv`. Figure 3 uses the mean and odd-median columns of `support.csv` and `mass.csv`, writing the exact derived curves to `mixed_cost.csv` and target crossover weights to `mixed_crossovers.csv`. `generated/numbers.tex` supplies narrative empirical numbers.

In storage only, infeasible K^* is encoded as $d + 1$, never as a measured finite investment; infeasible mass is infinity. All reported mean support costs use $\rho = 1$, where the cohorts are feasible. Partial-response figures use feasibility indicators. `witnesses.json` contains selected target/task lists and before/after scores. `witness_audit.json` records the full audit count. The tests compare all four support solvers against exhaustive search with zeros, ties, and nonuniform caps; independent LP and constrained-optimization checks cover continuous median and geometric costs. Proof constructions are tested as well, but never included as empirical data.

F Completed reporting and artifact checklist

The inspected ICLR 2027 author guidelines and official template require an AI use statement and recommend ethics and reproducibility statements. Those are completed above. They do not prescribe a separate NeurIPS-style paper checklist. The following additional checklist documents this deliverable's claims and limits.

1. **Claims bounded by evidence: Yes.** Every empirical statement concerns the frozen panels and declared gain boxes. No training-cost or misconduct inference is made (Sections 3 and 6).
2. **Assumptions and proof coverage: Yes.** Bounds, no spillovers, fixed rivals, ties, zeros, feasibility and even-median handling are explicit. Full proofs appear in Appendix A.
3. **Data provenance and selection: Yes.** Versions, URLs, checksums, precision priority, dropped records and licensing uncertainty are recorded in the artifact and Appendix B.
4. **Reproducible code and environment: Yes.** Pinned requirements, seeds, commands, exact score matrices, per-target results and scripts for every table and figure are included (Appendix E).
5. **Baselines and ablations: Yes.** Lowest-first, random order, partial responses, ceilings, score mass, alternative cohorts and normalization, positive margins and subject weights are reported.
6. **Uncertainty: Yes, within scope.** Paired owner reweighting and task perturbations are reported as sensitivity ranges. Test-item accuracy uncertainty and uncertainty in training responses are not estimated; the missing information is stated explicitly.
7. **Computational resources: Yes.** CPU time and peak memory are logged. No GPU training, model evaluation or paid experimental API was used.
8. **Negative and unanticipated results: Yes.** Infeasibility, task-scale dependence, task-resampling sign changes and the score-mass reversal are retained.
9. **Human subjects or private data: Not applicable.** Only public aggregate scores are analyzed. No privacy-sensitive examples are collected.
10. **AI assistance and human review: Disclosed.** All research and writing were agent-produced; no independent human review is claimed.
11. **Restricted access protocol: Followed.** Only the allowed starting papers, a directly addressed older election-bribery paper, and official data, software metadata and template/policy pages were consulted. No forward/backward citation search or attempt to identify the excluded paper was performed. The access log reports no recognized incidental encounter with excluded material.
12. **Submission status and format: Complete report, not submitted.** Official ICLR 2027 files and the nine-page initial main-text limit are used; there is no claim of peer review or acceptance.